

RF-based Motion-sensing Robot

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We have seen many projects where a robot is controlled by a remote controller. Here we present a robot that can be moved in any direction just by the movement of your hand. This motion-sensing robot senses the motion of your hand and transmits corresponding instructions to control the robot through RF communication.

Fig. 2 shows block diagram of the motion-sensing robot. The movement of wireless controller is sensed by the

accelerometer. The data from the accelerometer is compared by the comparator and corresponding encoded instructions are transmitted through the RF transmitter to control the robot. The robot receives the instructions and the microcontroller section drives the motors corresponding to the instructions received.

Circuit and working

Figs 3 and 4 show circuits of the wireless controller and the robot, respectively.

The wireless controller is built around encoder HT12E (IC1), operational amplifier LM339 (IC2), regulator 7805 (IC3), RF transmitter module (TX1), accelerometer module and a few

discrete components.

IC2 has four op-amps that are wired as comparators. Pins 9 and 10 of IC2 are connected to X-axis output of the accelerometer. Pins 4 and 7 are connected to Y-axis output of the accelerometer. IC2 receives the data for X and Y axes from the accelerometer. This data is continuously compared with pre-defined values for each axis by comparators A1 through A4. The reference value is selected using VR1 through VR4.

If the received signal for any axis

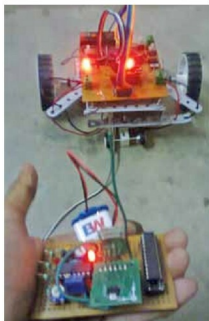


Fig. 1: Author's prototype

Test Points for Fig. 2

Test point	Details
TP0, TP4	Ground (GND)
TP1, TP5	5V
TP2	Input from Y-axis
TP3	Input from X-axis
TP6	0V when S3 is pressed
TP7	Input for PB0 of IC6 (D1)
TP8	Input for PB1 of IC6 (D2)
TP9	Input for PB2 of IC6 (D3)
TP10	Input for PB3 of IC6 (D3)

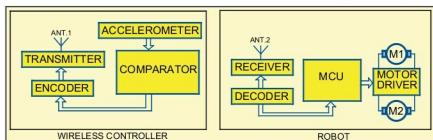


Fig. 2: Block diagram of motion-sensing robot

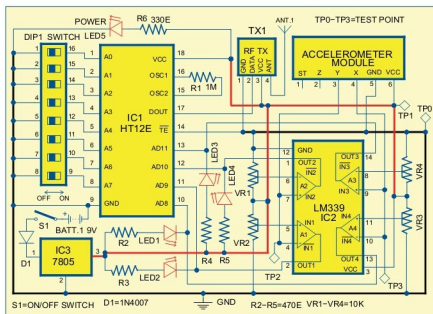


Fig. 3: Circuit of wireless controller